

Project for AI Engineer Role at Madeline & Co.

Vision-Language Report Generator for Business Analytics

Overview

Build an **AI-powered backend system** that generates structured business reports from mixed data sources — tabular data (CSV) and visual data (images such as charts or infographics).

The system should use **vision-language models** to interpret visuals, **LLMs** to analyze tabular data, and **LangChain agents** to synthesize insights.

The final output must include **summaries, metrics, and trend analysis**, returned as both **JSON** and **optional downloadable PDF reports**.

This project tests applied understanding of **multimodal AI, prompt engineering**, and **backend architecture design** using production-ready tools.

Tech Stack

- **Backend Framework:** FastAPI
 - **AI & Orchestration:** LangChain, LLMs (OpenAI / Gemini / Local), CLIP / BLIP / Any Vision Model
 - **Database:** PostgreSQL
 - **Vector Store:** Qdrant
 - **Storage:** AWS S3
 - **Deployment:** Any cloud host (Render / Railway / AWS / Hugging Face Spaces)
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Core Features (Required)

1. Data Ingestion

- Accepts multiple **CSV** and **image** files via API endpoints.
- Stores metadata in PostgreSQL and embeddings in Qdrant.
- Uploads original files to S3 (or mock equivalent).

1. AI Processing Pipeline

- Uses **Vision Model (CLIP/BLIP)** to extract information from images.
- Uses **LLM + LangChain Agent** to analyze CSVs and combine insights.
- Generates structured output with:
 - Key metrics
 - Trends and correlations
 - High-level recommendations
 - Summary paragraph

1. Report Generation

- Returns final structured report as **JSON** via API.
- Optionally generates **PDF version** of the report with:
 - Headings, tables, charts (if possible)
 - Proper formatting (bonus points)

1. API Documentation

- Fully documented **Swagger UI** generated via FastAPI.
- Clear descriptions for all routes, request/response models, and parameters.

1. Deployment

- Working API deployed publicly (e.g., on Render / Railway / AWS EC2 / Hugging Face Spaces).
 - Live endpoint accessible for testing.
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Bonus / Optional Features

- Generate **embeddings** and perform similarity search across documents.
 - Add **simple authentication** for API access.
 - Include **unit tests** for major modules.
 - Enhanced **PDF styling** with embedded charts and tables.
 - Integrate a **logging system** or error-handling middleware.
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Suggested Use Case (Optional)

Retail Sales Insights Generator

The model analyzes CSVs of monthly sales and marketing campaign images (e.g., posters, charts) to produce a “Business Summary Report” with insights like:

- Monthly performance metrics
- Visual interpretation of campaign results
- Trend summary and next-step recommendations

Candidates may choose this use case or propose their own, but it must follow the same multimodal input structure.

Evaluation Criteria

Category	Description	Weightage
Code Quality & Modularity	Clean, readable, well-structured, modular code	20%
AI Integration	Proper and meaningful use of LLMs + Vision models	25%
Architecture & Design	Logical backend design, proper use of LangChain and storage layers	25%
Documentation & API Clarity	Clear README, Swagger docs, setup instructions	15%
Deployment	Working, accessible endpoint with correct outputs	10%
Bonus Features	Test cases, PDF styling, or advanced insights	5%

Submission Guidelines

1. Submit via **GitHub repository** containing:
 - Complete codebase
 - **README.md** with setup, environment, and usage instructions
 - Example request/response JSON samples
 - (Optional) short Loom/video demo (≤ 3 min)
 1. API must be **live and accessible** at submission time.
 2. Deadline: **3 days** from assignment date (11:59 PM IST on Day 3).
 3. All files and datasets must be self-sourced — no pre-provided inputs.
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Deliverables Checklist

- FastAPI project with working routes
- AI pipeline (Vision + LLM + LangChain integration)
- JSON output of structured report
- Optional PDF generation endpoint
- Deployment link
- GitHub repository + README + Swagger docs